Project Title: PlantCare AI: Advanced Plant Disease Detection Using Transfer Learning

Project Design Phase-I - Solution Fit Template

Team ID:

CS 1. CUSTOMER SEGMENT(S)	TR 3. TRIGGERS	SL 10. YOUR SOLUTION
• Small-scale vegetable and crop farmers • Enthusiastic home and urban gardeners	• Spotting yellowing leaves, powdery mildew, or strange black spots on crops • Noticing a sudden drop in crop yield or sudden wilting of garden plants	• A web-based AI diagnostic tool • User takes a photo of the affected leaf and uploads it to the Flask web app • The underlying CNN analyzes the image to instantly identify the specific disease out of 38 categories
J&P 2. JOBS-TO-BE-DONE / PROBLEMS	**EM 4. EMOTIONS: BEFORE / AFTER**	**AS 5. AVAILABLE SOLUTIONS**
• Quickly and accurately identify the disease affecting their plants • Figure out the correct treatment to apply • Save crops from dying and prevent the disease from spreading	**Before:** Anxious about losing money, frustrated by conflicting advice, confused by symptoms **After:** Relieved to have an answer, confident in applying the right treatment, in control of their farm/garden	• Searching Google Images for matching leaf symptoms (often inaccurate) • Asking neighbors or staff at local agricultural supply stores • Trial-and-error spraying of broad-spectrum chemical pesticides
CC 6. CUSTOMER CONSTRAINTS	**BE 7. BEHAVIOUR**	**RC 9. PROBLEM ROOT CAUSE**
• Lack of formal botanical or agricultural education • Limited budget to hire professional agronomists • Sometimes limited internet bandwidth in rural farming areas	• Plucking the diseased leaves off manually • Taking physical samples of the plant to a local store for analysis • Delaying treatment because they are unsure what to do	• Plant diseases are visually very similar (e.g., various types of blight or rust) and require specialized scientific knowledge to differentiate • This expert knowledge is not democratized or easily accessible to everyday growers

CH 8. CHANNELS OF BEHAVIOUR	
8.1 ONLINE	**8.2 OFFLINE**
• Google searches for plant disease symptoms • Farming Facebook groups and communities • YouTube gardening and farming tutorials	• Local farming community meetings • Agricultural chemical shops for advice • Local markets and farmer networks